CSE 594 Assignment 2: AI-Assisted Used Car Selection

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Part 1: Identify or design the task

1.1 The task in the real world

The broad task is to help a person **buy a car**, or at least make a more concrete, better-informed choice when looking at several cars under their own constraints and background. This is a decision many people face, often more than once, and it is expensive, high-stakes and personal.

Most buyers are not experts. They may not know what resources exist (vehicle history reports, safety ratings, recall lookups), what the legal and contractual pitfalls are (branded or "washed" titles, dealer add-ons and fees that only appear in the fine print, as-is sales), or what ownership will cost later in maintenance, parts, insurance and warranty coverage. A listing that looks like a bargain today can be expensive for years. For example, a luxury car that cost around $250,000 new in 2016 may sell for $70,000–80,000 ten years later, but its parts and specialist service still cost what they did when it was new, which many buyers overlook. Some listings are simply *too good to be true*: the low price hides flood damage, a rebuilt odometer or a scam.

So, the job is not a yes/no classification of "buy this car or not". It is reasoning over the buyer's constraints, what they want, need, or may need in the future, and what each listing actually implies, then turning that into a choice the buyer understands. An AI assistant can do a large part of the analysis: check each car against every requirement, catch hidden problems, and explain what a detail means for the buyer now and later. The decision itself stays with the human.

1.2 Why neither the AI nor the human should do it alone

Why the human benefits from the AI:

- **Volume and diligence:** Even a small comparison means many listings, each with a dozen fields and a free-text note, checked against several requirements. People under time pressure skip checks; the AI checks every one.
- **Arithmetic:** All-in price (price plus fees) against the budget, mileage limits, and comparing qualifying cars on two priorities are tedious for people and quick for a model.
- **Knowledge many buyers lack:** Most first-time buyers don't know that a re-registered "clean" title after a total loss (title washing) or a request for a gift-card deposit are well-known warning signs. The AI names them explicitly.

- **Time and anxiety:** Researching a car takes time many people don't have, and buying one is stressful. An assistant that has already done the checking lets the buyer spend their attention on the decision rather than on the paperwork.

Why the human should not hand over the decision:

- **The AI does not know the person:** It has no inherent understanding of the buyer's needs, past experiences, current constraints, tastes, or plans for the future beyond what they tell it. For many people, and especially car enthusiasts, buying a car is an intimate decision that depends on how a car feels on a test drive, trust in the seller, appetite for risk, and preferences that change once they see the options.
- **The AI still makes confident mistakes:** My fine-tuned model is wrong on 14 of 200 study trials, and its wrong explanations sound exactly as sure as its right ones: "All-in price $10,199 is over the $22,500 budget" (Section 3.3).
- **People read everyday language well:** Several of the model's errors are human-obvious: "Some silt behind the dash panels from when the river crested" is flood damage, while "Not a scam, I promise! Happy to meet at the police station" is a careful seller, not a scam.
- **Consequences and accountability:** A wrong choice costs thousands of dollars and years of ownership. The person who pays and drives should own the decision.

Whether a decision is ultimately "right" for a person is subjective. The goal of the assistance is to equip the buyer with accurate, applicable information so they can make an educated choice, not to make the choice for them. In short, the AI is a fast, thorough checker that a human should verify, not an agent that should buy the car.

1.3 The task

With the time and resources available for this assignment, I want to study a narrower version of this task that can be completed in a few minutes and scored objectively:

> *Given a buyer's description of their situation and four used-car listings, recommend the listing that best meets the buyer's needs, or None if no listing does.*

Each trial takes about 2–3 minutes, so a participant can complete 3–4 trials in a 10-minute session. The buyer description includes an all-in budget, hard requirements (some stated outright, some only implied by the story, such as "I never learned to drive stick"), and two soft priorities. Each listing has the usual structured fields plus a free-text seller note.

What makes it non-trivial is that the decisive information is often not in the structured fields. A clean-title car whose seller note says it sat in flood water, a dealer whose fine print adds $2,000 in mandatory fees, or a seller who wants a gift-card deposit before any viewing all fail the buyer's requirement that the car be honestly represented. These are exactly the too-good-to-be-true situations from the real task.

What is in and out of scope: I have kept the parts of the real task that can be checked objectively, including: requirements, budget including fees, hidden problems in seller notes, needs

implied by the buyer's story, and trade-offs between qualifying cars. The scored answer is the requirement-checking decision. I left out long-term ownership-cost estimates (maintenance, parts, insurance) and live lookups of recalls, safety ratings or dealer reviews; my model does not browse the web. The subjective side (how the car feels, trust, taste) is intentionally not scored. In Assignment 3 I will collect it separately: participants' confidence, "would you actually buy this?", and a short reason.

The exact instructions participants will see, what they are given, and a worked example trial are in the final section, Study guidelines and participant expectations.

Part 2: Identify or create a model to provide AI assistance

2.1 Dataset: my own task and dataset (bonus)

File names in this report link to the GitHub repository; the same files are in the `car_advisor/` folder submitted on Canvas. I designed the task and created all data myself with my generator (`gen_dataset.py`, fixed random seeds). No existing dataset was sampled. I considered a public Craigslist used-car listings dataset (Zenodo record 10457828, https://zenodo.org/records/10457828) but did not use its rows, because real listings come without a ground truth for "which car should this buyer pick". The make and model names are real vehicles, offered only in model years they were actually produced, but every price, mileage figure, seller and note is fictional. Table 1 lists the three splits I generated and what each one is used for; the reasons for their sizes follow.

Table 1: The three data splits. Each trial has one buyer and four listings.

Split	Trials	Listings	Used for
Study	200	800	Evaluation (Part 3) and Assignment 3 user study; excluded from model training and checkpoint selection
Training	1,620	6,480	LoRA fine-tuning
Validation	180	720	Choosing the training checkpoint (lowest validation loss)

Why these sizes:

- **Study set, 200 trials:** This meets the assignment's minimum and is large enough for balanced strata: 30/70/70/30 trials across the four trial types and 40 per buyer type. It also matches the Assignment 3 study: with an assumption of approximately 70 classmates doing 3–4 trials each (roughly 210–280 completed trials), a larger set would leave most trials with no human answers at all.
- **Training and validation, 1,800 trials split 90/10:** I wanted enough examples for the model to learn the output format and the checking rules across all buyer types, requirements and red-flag kinds, while keeping training to minutes on one GPU. The training set uses the same mix of trial types as the study set (35% trap, 35% trade-off, 15% clear, 15% none). The validation loss flattens over the second epoch (Figure 1), which suggests more data of the same kind would add little. I did not run a formal data-size ablation.

Table 2: Contents of one trial.

Part	Field	Description
Input (shown)	Buyer profile	A first-person story in plain language: who the buyer is, their all-in budget, their requirements (stated or implied), the honest-representation requirement, and their two priorities
	Listings A–D	Year, make, model, price, mileage, body style, seats, drivetrain, transmission, fuel economy (mpg), title status, seller type, location, and a free-text seller note
Hidden: buyer	Buyer type	One of five (first-time, commuter, family, cold-weather, outdoor/lifestyle)
	Budget	All-in limit in dollars, including required fees
	Requirements	Automatic, AWD/4WD, maximum miles, minimum seats, body type; each marked *stated*, *implied*, or not required
	Priorities	Two of: lower price, lower mileage, newer year, better fuel economy, in order
Hidden: listings	Required fees	Dealer fees that must be paid, including those mentioned only in the note
	Violations	Each requirement the listing breaks and where it is visible: in the fields (*explicit*), only in the buyer's story (*implied*), or only in the seller note (*note*)
Hidden: answer key	Feasible options	Listings that meet every requirement
	Acceptable choices	Feasible listings not beaten on both priorities, or None
	Best choice	The single top pick under a 0.6 / 0.4 weighting
	Trial type, wording	Clear, trap, trade-off or no suitable car; generated or hand-written

What a trial contains: Every trial in every split has the same structure (Table 2). Participants and the model only see the input; the structured requirements and the answer key are hidden and used for scoring. The submitted CSV has one row per trial with these fields, plus the model's output and its evaluation.

Why is it built it this way:

- **Generated, not real listings:** Real listings have no ground truth for which car a particular buyer should choose. Generating trials gives every trial a known correct answer, lets me control difficulty and balance, and avoids scraping listing sites against their terms of use or exposing real sellers.
- **Four listings per trial:** Allows for a real comparison with trade-offs, while still fitting into 2–3 minutes so participants can do several trials in a 10-minute session.
- **Tempting hidden-problem cars:** A car with a hidden problem always beats every qualifying car on the buyer's top priority. Otherwise, it could be ruled out without reading the seller note, and the trial would not test the skill that matters most in real car buying.
- **Stated and implied requirements:** Real buyers describe their lives ("my doctor said no clutch work for a year"), not checklists. Mixing both forms enables us to test whether the performer understands the story rather than matching keywords.

- **The honest-representation requirement:** Every buyer says the car must be honestly represented. That makes flood or frame damage, washed titles, unreliable odometers, scams and looming major repairs objective deal-breakers instead of matters of taste, so they can be scored.
- **No-suitable-car trials:** Sometimes the right answer is to walk away. These trials test whether the performer, or the AI, notices when nothing qualifies, and whether people follow an AI that recommends a car anyway.
- **A rule-based answer key:** The correct answer is computed from the structured data, not judged by me. It is objective and reproducible, and it scores the AI and human participants in exactly the same way.

How a trial is built:

1. **Buyer:** One of five buyer types (first-time buyer, daily commuter, family, cold-weather driver, outdoor/lifestyle) sets a budget range ($7,000–$36,000) and likely requirements. Each requirement is randomly phrased outright or implied in the story. Every buyer also states that the car must be honestly represented.
2. **Listings:** Cars are sampled from a 38-model catalog with real production years. Price is derived from age, mileage and a new-car price, with noise.
3. **Roles:** Each listing is assigned a role: *qualifies*, *visible violation* (for example over budget or a salvage title), or *hidden problem* (a deal-breaker only in the seller note or only implied by the buyer's story). Hidden-problem cars are deliberately tempting: they beat every qualifying car on the buyer's top priority.
4. **Answer key:** Computed by rule from the trial's structured data (Section 3.1 gives the scoring rules).

The generator mixes these roles into four trial types, each testing a different skill (Table 3). Most of the study set is trap and trade-off trials, because those are the ones where reading the fine print and weighing priorities matter.

Table 3: Composition of the 200 study trials.

Trial type	Trials	What it tests
Clear	30	One qualifying car; every other car fails visibly
Trap	70	One qualifying car plus a tempting car with a hidden problem
Trade-off	70	Two or three qualifying cars that trade off the buyer's priorities
No suitable car	30	Every car fails; the right answer is None

The trials are balanced across the five buyer types (40 each) and the correct letter (A 40, B 44, C 48, D 38, None 30). In 116 trials at least one listing has a hidden problem. Across all 800 listings there are 122 seller-note deal-breakers: flood damage (16), frame damage (26), washed title (16), unreliable odometer (17), scam-like payment request (18), major repair needed (14) and required fees that break the budget (15). There are also 32 implied-need violations.

Separation of datasets: The training and validation sets come from a different seed. They contain do not contain any of the study trials, buyer texts or car sets (checked programmatically), and not a single one of the study-only hand-written phrasings described below.

Dataset iterations:

- **Iteration 1 to 2: hand-written wording:** In the first version of the study set, every seller note and implied need came from the same kinds of templates as the training data. The fine-tuned model scored 92.5% on it, which raised two concerns: the score might reflect memorized phrasing rather than understanding, and a near-perfect AI on template text would make the task one people should simply delegate. Thus, I rewrote 80 of the 200 study trials (40%, stratified by trial type) with hand-written notes and buyer stories that never appear in training: red flags buried in otherwise positive notes, harmless notes with alarming words ("I rebuilt the front suspension myself"), and implied needs told as anecdotes. The answer key is unchanged, so the two versions are directly comparable (Section 3.4). This revision was guided by the study-set result, as the assignment's iteration step allows; the model was not retrained or tuned in response, and the change made the study set harder for the model, not easier.
- **Data realism fix:** While preparing the report I found that the generator could pair a model with a year in which it was never made (for example, a 2015 Kia Telluride, which launched in 2020). Hence, I added real production-year ranges, regenerated all splits from the same seeds, retrained the model and reran every evaluation. All numbers in this report are from the regenerated data. The results before the fix were very similar (base 72.0%, fine-tuned 92.5%), so no conclusion depends on it.

2.2 The AI assistance

For every trial the model returns JSON with five parts, which the Assignment 3 interface will show next to the listings:

1. `option_checks`: a one-line verdict for each listing ("Fails: required dealer fees bring the all-in price to $19,894, over the $19,000 budget").
2. `red_flags`: problems found in seller notes, with the listing and the issue.
3. `recommendation`: A, B, C, D or NONE.
4. `reason`: why that listing best fits the buyer.
5. `tradeoffs`: what the buyer gives up compared with the other qualifying listings.

The per-listing checks and red flags let a human verify the AI's work quickly instead of taking the recommendation on trust.

2.3 The model and why I chose it

The assistant is **Qwen3-4B-Instruct-2507** [1], **fine-tuned by me with LoRA** [2] on my 1,620 training trials. I chose it because:

- **Open weights (Apache-2.0):** I can fine-tune it and host it myself, so the Assignment 3 interface can show live AI output instead of copied chat-interface answers.
- **Small enough to train and serve cheaply:** At 4 billion parameters it fine-tunes in minutes on one GPU and answers a trial in about a second when batched, which matters for an interactive study.
- **Strong instruction following for its size**, including reliable JSON output, and the 2507 "Instruct" variant answers directly without long hidden reasoning traces, which keeps responses fast and predictable.

Initial performance and why I fine-tuned: With a careful prompt, the base model chose an acceptable car on 72.5% of study trials. Its errors were systematic (Figure 4a): it often took the tempting car with a hidden problem (it flagged only 86.9% of seller-note deal-breakers), picked cars beaten on both priorities, said None when a car qualified, raised false alarms on 16.2% of fully qualifying cars, and once produced unparseable output. That is likely no better than an attentive person, so it would be weak assistance. Fine-tuning targets those behaviors: checking every listing against every requirement, telling deal-breakers from cosmetic issues, and abstaining only when nothing qualifies.

How I fine-tuned: Each training example is a buyer profile and four listings (input) and a reference answer in the same JSON format (target). Reference answers are generated from the answer key: the true violations of each listing, the hidden red flags, the correct choice, and a numeric comparison of the qualifying alternatives. I trained with TRL's `SFTTrainer` (`train_lora.py`): LoRA rank 16, alpha 32, dropout 0.05, on all attention and MLP projections (33M trainable parameters, 0.81% of the model); 2 epochs, learning rate 10^{-4} with cosine decay and 20 warm-up steps, effective batch size 8, bf16, maximum length 3,072 tokens, loss on the answer tokens only. The checkpoint with the lowest validation loss is kept; it was the final step (validation loss 0.006, Figure 1). Inference uses greedy decoding, so outputs are deterministic, and base and fine-tuned models get the identical system prompt and input.

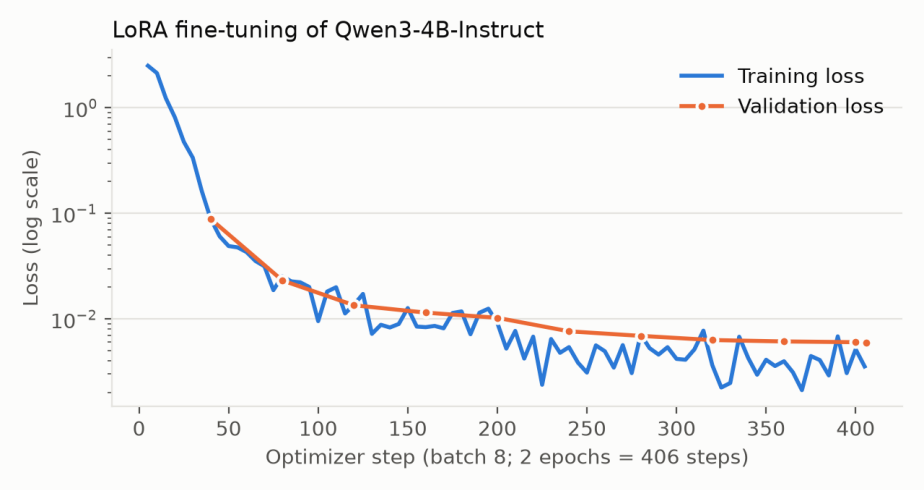


Figure 1: Training and validation loss during LoRA fine-tuning (log scale).

What fine-tuning changed: The acceptable-choice rate rose from 72.5% to 93.0%, red-flag recall from 86.9% to 99.2%, and false alarms fell from 16.2% to 6.4% (full comparison in Section 3.2). Not everything improved: performance on clear trials did not change, the model scores a few points lower on hand-written wording than on generated wording, and it became over-cautious, so that wrongly saying None is now its most common outcome error (Section 3.3). Figure 2 shows the fine-tuned model running on one study trial, as evidence that it works end to end.

2.4 Replicating and hosting the model

What someone would need to do: The model is not hosted anywhere public, so anyone who wants to use it runs it on their own machine. The code, data and saved outputs are in the `car_advisor/` folder submitted on Canvas and in the repository https://github.com/rajdeeepm/ai-car-advisor. The trained LoRA adapter is always in the repository, and in the Canvas zip too if that version includes it, so retraining is optional. The adapter (130 MB) is over GitHub's 100 MB file limit, so it is stored as two zip parts; `setup.sh` joins them, checks them against a checksum and unzips the adapter into `model/lora_adapter/`:

1. Get the adapter. The full Canvas zip has it already unzipped in the `model` folder. For the smaller Canvas zip, which leaves it out, clone the repository and run `bash setup.sh`; a note in that zip gives the same steps.
2. Install Python 3.12 with PyTorch 2.13 (CUDA), Transformers 5.17, PEFT 0.21 and TRL 1.15 (the versions I used; `requirements.txt` and the `README` list them).
3. Download the base model from Hugging Face (`Qwen/Qwen3-4B-Instruct-2507`, about 8 GB).
4. **To use my model:** run `infer.py` (a whole split) or `demo.py` (one trial), passing the adapter folder with the `-lora` option. For a live interface, the same loading code can sit behind a small web endpoint; a serving engine such as vLLM can make it faster.
5. **To retrain it:** regenerate the data with `gen_dataset.py` and run `train_lora.py`, or run `run_all.sh`, which trains and then produces every prediction used in this report.

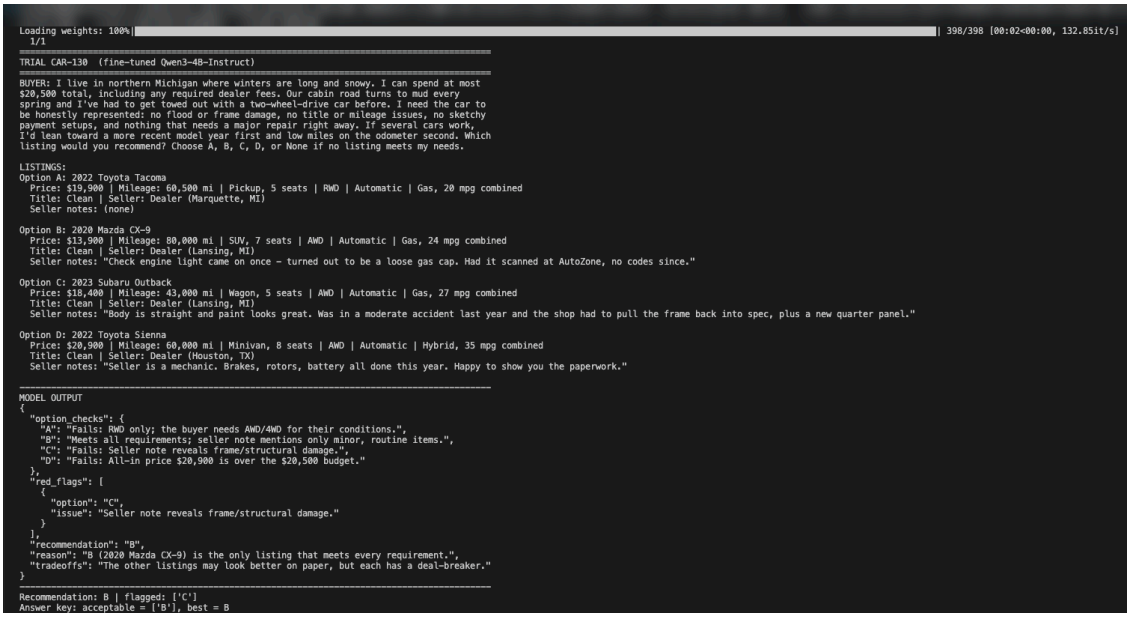
Resources needed: One NVIDIA GPU with CUDA. About 24 GB of GPU memory is enough for inference. Training also runs on one GPU; on cards with less memory, lower the per-device batch size and raise gradient accumulation to keep the effective batch at 8 (for example on a Google Colab A100). About 10 GB of disk covers the base model, adapter and data.

Part 3: Evaluate the model

3.1 Benchmarking: method and metrics

How I benchmarked the model performance: Both models, the base Qwen3-4B-Instruct and my fine-tuned version, answer all 200 study trials with the identical prompt and greedy decoding. Each output is parsed and scored automatically against a rule-based answer key (`evaluate.py`):

1. A listing is **feasible** only if it meets every requirement: the all-in budget (price plus required dealer fees), no branded title, any required transmission, drivetrain, seats, body type or mileage limit (stated outright or implied in the story), and no deal-breaker in the seller note




```
TRIAL CAR-130  (fine-tuned Qwen3-4B-Instruct)
BUYER: I live in northern Michigan where winters are long and snowy. I can spend at most
$20,500 total, including any required dealer fees. Our cabin road turns to mud every
spring and I've had to get towed out with a two-wheel-drive car before. I need the car to
be honestly represented: no flood or frame damage, no title or mileage issues, no sketchy
payment setups, and nothing that needs a major repair right away. If several cars work,
I'd lean toward a more recent model year first and low miles on the odometer second. Which
listing would you recommend? Choose A, B, C, D, or None if no listing meets my needs.

LISTINGS:
Option A: 2022 Toyota Tacoma
  Price: $19,900 | Mileage: 60,500 mi | Pickup, 5 seats | RWD | Automatic | Gas, 20 mpg combined
  Title: Clean | Seller: Dealer (Marquette, MI)
  Seller notes: (none)
Option B: 2020 Mazda CX-9
  Price: $13,900 | Mileage: 80,000 mi | SUV, 7 seats | AWD | Automatic | Gas, 24 mpg combined
  Title: Clean | Seller: Dealer (Lansing, MI)
  Seller notes: "Check engine light came on once - turned out to be a loose gas cap. Had it
                 scanned at AutoZone, no codes since."
Option C: 2023 Subaru Outback
  Price: $18,400 | Mileage: 43,000 mi | Wagon, 5 seats | AWD | Automatic | Gas, 27 mpg combined
  Title: Clean | Seller: Dealer (Lansing, MI)
  Seller notes: "Body is straight and paint looks great. Was in a moderate accident last year
                 and the shop had to pull the frame back into spec, plus a new quarter panel."
Option D: 2022 Toyota Sienna
  Price: $20,900 | Mileage: 60,000 mi | Minivan, 8 seats | AWD | Automatic | Hybrid, 35 mpg combined
  Title: Clean | Seller: Dealer (Houston, TX)
  Seller notes: "Seller is a mechanic. Brakes, rotors, battery all done this year. Happy to
                 show you the paperwork."

MODEL OUTPUT
{
  "option_checks": {
    "A": "Fails: RWD only; the buyer needs AWD/4WD for their conditions.",
    "B": "Meets all requirements; seller note mentions only minor, routine items.",
    "C": "Fails: Seller note reveals frame/structural damage.",
    "D": "Fails: All-in price $20,900 is over the $20,500 budget."
  },
  "red_flags": [
    { "option": "C", "issue": "Seller note reveals frame/structural damage." }
  ],
  "recommendation": "B",
  "reason": "B (2020 Mazda CX-9) is the only listing that meets every requirement.",
  "tradeoffs": "The other listings may look better on paper, but each has a deal-breaker."
}
Recommendation: B | flagged: ['C']
Answer key: acceptable = ['B'], best = B
```

Figure 2: The fine-tuned model on study trial CAR-130 (`demo.py`): screenshot (top, machine details blurred) and the same output as text (bottom). C hides frame damage in its note, A is rear-wheel drive (ruled out by the muddy cabin road), and D is $400 over budget. The model flags C and recommends B, matching the answer key.

(flood or frame damage, washed title, unreliable odometer, scam-like payment request, major repair needed).

2. If no listing is feasible, the correct answer is **None**.
3. Otherwise a choice is **acceptable** if it is feasible and no other feasible listing is at least as good on both of the buyer's stated priorities and strictly better on one (it is Pareto-optimal on the two priorities). The single **best** choice uses a 0.6 / 0.4 weighted score of the two priorities, each min-max normalized within the trial.

As a sanity check, the reference answers used for training score 100% under these rules.

Metrics and why I chose them:

- **Acceptable-choice rate (primary):** A buyer who says "low miles first, then price" has not told me how many dollars a mile is worth, so when two cars each win on one priority, either is a defensible recommendation. Scoring only an exact match would penalize reasonable answers for my own arbitrary weighting. This is also the metric I will use for humans in Assignment 3, so all three conditions are compared on the same scale.
- **Exact best choice (secondary):** Agreement with the 0.6/0.4 weighted pick, reported for completeness.
- **Meets all requirements:** The share of trials where the model did *not* recommend a car that breaks a hard requirement or wrongly say None. These are the expensive errors for a real buyer (a flood car, a scam), as opposed to picking a slightly worse qualifying car.
- **Red-flag recall and false-alarm rate:** The interface shows the AI's red flags, so their quality matters on its own: recall is the share of seller-note deal-breakers the model flagged; the false-alarm rate is the share of fully qualifying listings it flagged anyway.
- **None precision and recall:** How reliably the model abstains when nothing qualifies, and only then.

How the metric choice affects the results: The choice matters most for trade-off trials. The fine-tuned model's acceptable-choice rate (93.0%) is higher than its exact-best rate (86.5%): in the trials in between, it picked a car that is defensible for the buyer but not the one my particular weighting prefers. Scoring by exact match would make the AI look worse for reasons a buyer would not care about, and it would make the human–AI comparison in Assignment 3 depend on an arbitrary weighting. The separate red-flag and requirement metrics show *how* the model is right or wrong, which matters because the interface shows its reasoning, not just its answer.

3.2 Results: before and after fine-tuning

Fine-tuning raised the acceptable-choice rate from 72.5% to 93.0% and improved every metric (Table 4, Figure 3), which justifies fine-tuning in the first place. Table 5 breaks the acceptable-choice rate down by trial type and by wording, to show *where* fine-tuning helped and whether it holds up on text it never saw in training. The largest gains are on the hard trial types: trap trials (72.9% to 92.9%), where the base model often took the tempting car; trade-off trials, where it often picked a car another qualifying car beats on both priorities (65.7% to 92.9%); and no-suitable-car

Table 4: Base vs. fine-tuned model on the final 200-trial study set. ↑: higher is better; ↓: lower is better (false alarms and unparseable outputs).

Metric	Base Qwen3-4B	Fine-tuned (LoRA)
Acceptable choice, primary (↑)	72.5%	93.0%
Exact best choice (↑)	66.5%	86.5%
Meets all requirements (↑)	81.0%	95.5%
Red-flag recall (↑)	86.9%	99.2%
Red-flag false alarms (↓)	16.2%	6.4%
None precision (↑)	61.8%	82.9%
None recall (↑)	70.0%	96.7%
Unparseable outputs (↓)	1	0

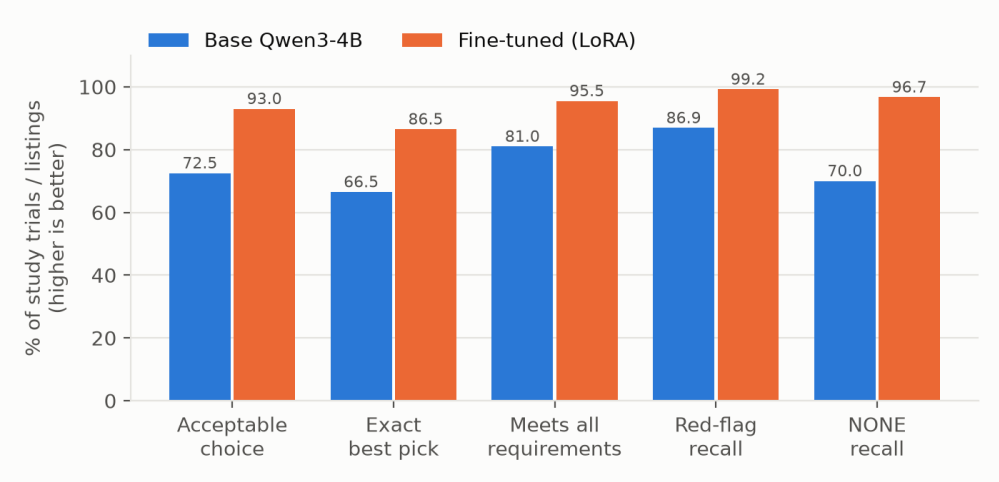


Figure 3: Before (base) and after (fine-tuned) on the main metrics. Higher is better for every metric shown.

Table 5: Acceptable-choice rate by trial type and by wording (higher is better).

Subset	Trials	Base	Fine-tuned
Clear	30	90.0%	90.0%
Trap	70	72.9%	92.9%
Trade-off	70	65.7%	92.9%
No suitable car	30	70.0%	96.7%
Generated wording	120	73.3%	94.2%
Hand-written wording	80	71.2%	91.2%
All	200	72.5%	93.0%

trials (70.0% to 96.7%). On hand-written wording, which never appears in training, the fine-tuned model scores 91.2% versus 94.2% on generated wording, so it generalizes to new phrasing with a small drop. Both models score the same on clear trials, where every wrong car fails visibly.

3.3 Error analysis

Following standard error-analysis practice [5], I first assign every wrong answer an outcome category automatically (`evaluate.py`): what went wrong. I then read each of the 14 fine-tuned errors by hand to find its root cause: why it went wrong (`results/ft_error_root_causes.csv`). Figure 4 shows both distributions.

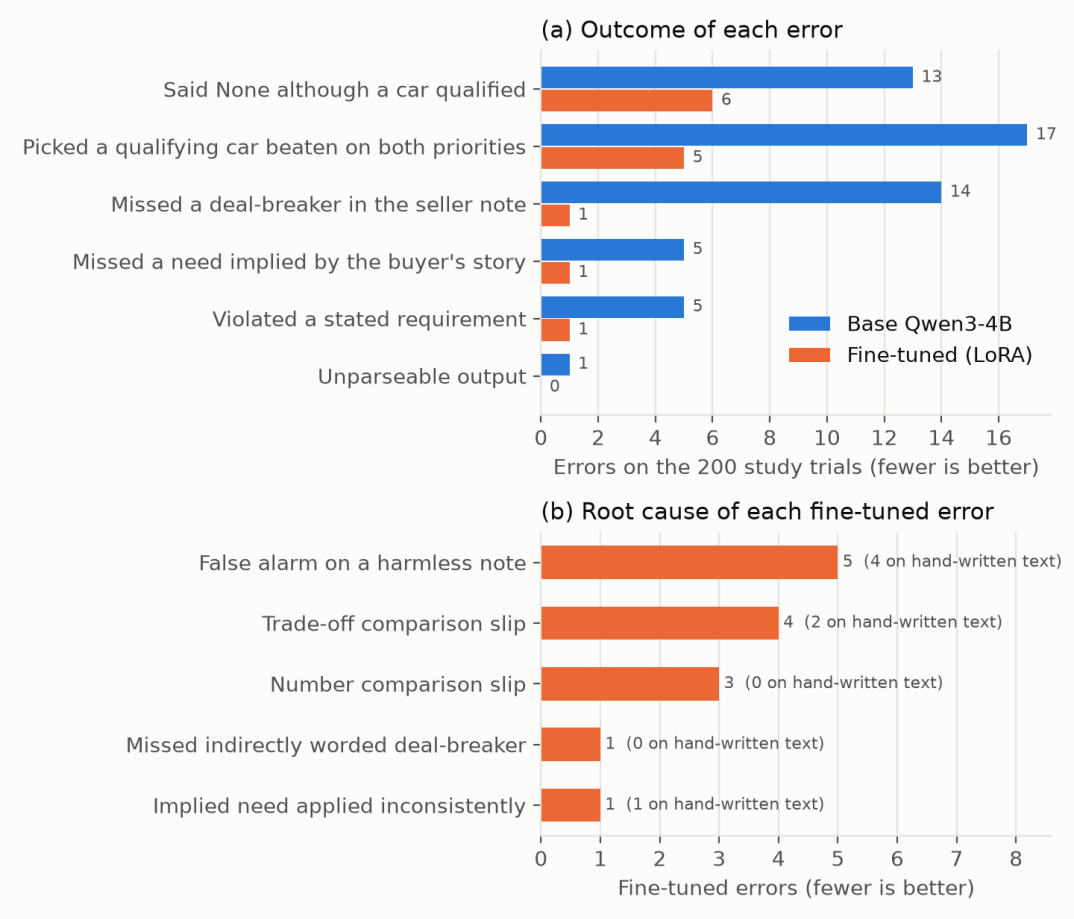


Figure 4: Error distributions. (a) Outcome category of every error, base vs. fine-tuned. (b) Hand-labeled root cause of each fine-tuned error, with how many occurred on hand-written wording. Bars count errors, so fewer is better.

Reading the error tables: Table 6 gives the exact counts behind Figure 4a. Fine-tuning cut errors from 55 to 14, and the mix changed: missed seller-note deal-breakers, a quarter of the base model's errors (14), fell to one, while unnecessary None answers became the fine-tuned model's largest error type (6 of 14). Table 7 gives the counts behind Figure 4b, with a real example of each root cause from the study set; the two most common causes, false alarms and trade-off slips, make

Table 6: Error distribution by outcome (automatic). Counts: lower is better. Percentages are shares of each model's errors and describe the error mix, not better or worse.

Error pattern	Fine-tuned	Base
Said None although a car qualified	6 (42.9%)	13 (23.6%)
Picked a qualifying car beaten on both priorities	5 (35.7%)	17 (30.9%)
Missed a deal-breaker in the seller note	1 (7.1%)	14 (25.5%)
Violated a stated requirement	1 (7.1%)	5 (9.1%)
Missed a need implied by the buyer's story	1 (7.1%)	5 (9.1%)
Unparseable output	0	1 (1.8%)
Total errors (of 200 trials)	14	55

Table 7: Root causes of the 14 fine-tuned errors (hand-labeled, one label per error). Counts: lower is better; percentages are shares of the 14 errors.

Root cause	Errors	% of errors	Example from the study set
False alarm on a harmless note	5	35.7%	"Not a scam, I promise! Happy to meet at the police station parking lot, bring your mechanic" was flagged as a seller demanding payment before any viewing (CAR-042); a DIY suspension rebuild was called "title washing" (CAR-192).
Trade-off comparison slip	4	28.6%	The model's own trade-off text says the alternative "has 2,000 fewer miles" in the same model year, the buyer's top two priorities, yet it recommends the other car (CAR-152).
Number comparison slip	3	21.4%	"All-in price $10,199 is over the $22,500 budget" (CAR-165); an all-in price of $11,699 called within an $11,500 budget (CAR-038).
Missed an indirectly worded deal-breaker	1	7.1%	"Some silt behind the dash panels from when the river crested" (flood damage) read as routine (CAR-059).
Implied need applied inconsistently	1	7.1%	The buyer's knee injury ("no clutch work") correctly ruled out one manual truck but not the other (CAR-180).
Total	14	100%	

up 9 of the 14 errors. The paragraphs below explain how each cause arises.

Overall pattern: Fine-tuning changed *how* the model fails. The base model mostly made permissive errors: it accepted the tempting car with a hidden problem or a dominated trade-off. The fine-tuned model rarely misses a deal-breaker (one miss in 200 trials) but has become *over-cautious*, and its remaining errors are small reasoning slips. The errors are spread across trial types (3 clear, 5 trap, 5 trade-off, 1 no-suitable-car). When it is wrong, it uses the same confident wording as when it is right ("Fails: ..." or "Meets all requirements").

How each error happens and likely causes:

- **False alarms on harmless notes (5):** The model matches alarming words ("fender-bender", "rebuilt", "scam", "pothole") to deal-breakers and sometimes invents a detail to justify it ("asks for payment before any viewing" is not in the note). Four of the five are on handwritten notes. A likely cause is that training taught a strong link from damage and payment vocabulary to "Fails", and its harmless notes rarely used those words, so the model never learned the benign readings. The training mix may add to this: in trap trials the most attractive car always hides a problem, which can teach the model that attractive cars are suspicious.
- **Trade-off comparison slips (4):** In every case the model's own trade-off text states that the other car is better on the buyer's top priority. A likely cause is the output order: the JSON asks for the `recommendation` before the `tradeoffs`, so the model commits to a choice before it writes out the comparison. Small differences (1 mpg, 2,000 miles) are also where small models are least reliable.
- **Number comparison slips (3):** The model computes the all-in price correctly but compares it with the wrong number, such as the car's own price instead of the budget, or reads $11,699 as within $11,500. These are typical of small language models, which handle numbers as text rather than calculating.
- **Indirectly worded deal-breaker (1):** "Silt behind the dash from when the river crested" requires an inference (silt + river = flood) that no training note phrased this way.
- **Inconsistent implied need (1):** The model checks each listing in a separate line with no consistency check across them, so the same inferred need (no clutch) was applied to one manual truck and not the other.

Can humans avoid these errors? Mostly yes, which is the basis for expecting complementarity:

- **False alarms (5)** are visible: the AI quotes a reason the human can check against the note in a few seconds, and in several cases the stated reason is simply not in the note. Its wrong None answers are also conspicuous: an AI that rejects all four cars invites a second look.
- **Comparison slips (7)** are visible in the AI's own output. "$10,199 is over the $22,500 budget" is wrong at a glance, and in all four trade-off errors the AI's trade-off text itself states that the alternative is better.
- **The two remaining misses** (flood silt, the knee injury) depend on reading ordinary language carefully. People are good at this when they read closely, though a rushed participant could miss them too.

The risk is the reverse: a participant who trusts the AI's confident tone will accept these errors, and one who distrusts it may override some of the 93.0% of trials it gets right. These are the over- and under-reliance effects studied in prior work [3, 4].

3.4 Adjustments for human–AI complementarity

What I changed and why: Two results drove my adjustments. First, the base model alone (72.5%) was too weak to be useful assistance, so I fine-tuned it (Part 2). I also made the output

Table 8: Acceptable-choice rate across study-set iterations (same trials and answer key; only the wording of 80 trials differs). Higher is better.

Study set	Base	Fine-tuned
Iteration 1 (generated wording only)	72.0%	92.5%
Iteration 2 (final, 40% hand-written)	72.5%	93.0%

parser tolerant of the base model's formatting quirks, so its score reflects its decisions rather than its formatting. Second, on the template-only iteration-1 study set the fine-tuned model scored 92.5%, close to the point where people should simply delegate, so I added hand-written wording to 40% of the study set (Section 2.1). The overall score barely moved (Table 8), but the model's remaining errors now cluster on exactly the kinds of text where human reading should help: alarming-but-harmless notes and indirect wording. That is what makes complementarity possible.

Why the study set is balanced, and why that matters after fine-tuning: The study set is not skewed toward one type of trial: 30 clear, 70 trap, 70 trade-off and 30 no-suitable-car trials; 40 trials per buyer type; correct answers spread across A–D and None (A 40, B 44, C 48, D 38, None 30); and 60% generated vs. 40% hand-written wording. This matters because the AI's and the humans' weaknesses sit in different trial types:

- People are likely to struggle on *trap* and *trade-off* trials, where the problem hides in fine print or requires arithmetic. The fine-tuned AI is strong there.
- The fine-tuned AI's errors are over-caution (wrong None answers, false alarms) and slips in close comparisons, which plain reading catches.

A set skewed toward clear trials would leave everyone near the ceiling, with nothing to complement; a set skewed toward traps would make the AI look nearly perfect and invite delegation. Balancing the answer letters also prevents participants or the model from scoring well by favoring one position, and including None trials tests whether people notice when the AI should, or should not, reject every car.

What I did not change: I considered making the task harder so the AI would score lower, but decided against it: the remaining AI errors are confident, varied and catchable by humans, which is the complementarity I want to test, and the task should keep a realistic mix of easy and hard trials.

3.5 Prediction

I predict: Human+AI > AI-only > Human-only, measured by how often the chosen car is an acceptable choice. The AI and the person make different kinds of mistakes, and each is good at what the other gets wrong, so together they should do better than either one alone.

- **Human+AI: highest, roughly 93–97%:** This range is an estimate that assumes participants actually check the AI's reasons against the listings; without that checking, Human+AI would land at or below the AI's 93.0% (see the risks below). The AI helps with what people tend to get wrong in a hurry: it adds up the fees, checks every requirement for every car,

compares the cars on the buyer's two priorities, and spots deal-breakers hidden in seller notes (it caught 99.2% of them). The person helps with what the AI gets wrong. My error analysis (Section 3.3) shows that 12 of the AI's 14 mistakes can be seen just by reading its answer. When it wrongly rejects a car, the reason it gives is often not in the note. When it compares numbers wrongly, the mistake is obvious ("$10,199 is over the $22,500 budget"). When it says no car works, that alone makes people look again. These mistakes happen mostly on harmless notes that sound alarming and on casual wording, which people read well.

- **AI-only: 93.0%** (measured, Table 4). Good, but it cannot catch its own mistakes: it sounds just as sure when it is wrong as when it is right.
- **Human-only: lowest, roughly 60–75%:** In 2–3 minutes per trial, a person has to check four cars against up to seven requirements, spot fees mentioned only in the fine print, work out needs the buyer only hints at, and compare the cars on both of the buyer's priorities to rule out any car that another one beats on both. The trap, trade-off and no-suitable-car trials are built around exactly the checks people skip when rushed, and many classmates may not know warning signs such as a "washed" title or a seller asking you to pay through an "eBay agent". I expect people to do well on clear trials and on obvious red flags, and worse on fees, hinted needs and trade-offs.

What could make this prediction wrong: The AI is already right on 93.0% of trials, so there is little room to beat it (only 14 mistakes in 200 trials). Past studies found that when the AI is much better than people, showing its reasons can make people simply agree with it more, instead of catching its mistakes [3, 4]. Human+AI only beats the AI alone if people fix more of its mistakes than they create by overruling its correct answers. Two things work against this: people trusting the AI when it is confidently wrong, and people ignoring the AI when it correctly rejects a car that looks like a good deal. To help, participants see the AI's reasons, not just its pick, so they can check each claim against the listing. Either way, Human+AI should do much better than a person alone.

Study guidelines and participant expectations (Assignment 3)

This section describes what participants will see and do in the user study.

What the participant is given:

- A **buyer profile** in plain language: who the buyer is, an all-in budget (price plus any required dealer fees), hard requirements, and two soft priorities (for example, "low miles first, then price"). Some requirements are stated outright ("It has to be an automatic"); others are only implied by the story ("I never learned to drive stick", "we have three kids and my parents ride along most weekends").
- **Four listings (A–D)**, each with year, make, model, price, mileage, body style, seats, drive-train, transmission, fuel economy, title status, seller type and location, plus a free-text **seller note**.
- In the **AI-assisted condition**, also the AI's recommendation, its check of each listing, the red flags it found, its reason and its stated trade-offs (Section 2.2).

What the participant must do: Choose A, B, C, D or None, and briefly explain why. After each choice I will also ask, without scoring it, how confident they are and whether they would actually buy that car. Each trial takes about 2–3 minutes, so a participant completes 3–4 trials in a 10-minute session.

How answers are scored: With the same rules as the AI (Section 3.1): a choice is correct if it meets every requirement and no other qualifying car is better on both priorities, or if it is None when nothing qualifies.

Planned trial selection: With approximately 70 classmates, I expect roughly 210–280 completed trials. Rather than spreading them over all 200, I plan to use a balanced subset of about 40–60 trials (all trial types, both wordings, and some of the AI's errors) so that each trial receives several answers in each condition and human-only and human+AI can be compared on the same trials.

Example trial CAR-131 (trap, hand-written wording), abridged

Buyer: "I work early shifts, so I'm often driving before the plows have been out. I can spend at most $25,000 total, including any required dealer fees. It needs all-wheel or four-wheel drive. I can't afford surprises, so no hidden damage, no questionable title or mileage history, no scam-like sellers, and no big repairs looming. Within those limits, what I care about most is getting something newer, and after that as few miles as possible."

	Listing	Seller note
A	2013 Hyundai Elantra, $2,500, 206,500 mi, FWD	"Check engine light came on once, turned out to be a loose gas cap."
B	2020 Subaru Ascent, $17,400, 43,500 mi, AWD	"Priced well below KBB because I need it gone before I leave the country Friday. eBay Motors protection: you pay through their agent and get 5 days to inspect after delivery."
C	2023 Toyota Sienna, $28,400, 22,000 mi, AWD	"Ate a pothole and needed a new tire and rim last winter."
D	2015 Honda Pilot, $7,400, 157,500 mi, AWD	"Water pump and timing belt were replaced at 95k as recommended."

Correct answer: D. A is front-wheel drive and C is over budget, both visible in the fields. B is the trap: it is the newest affordable AWD car, but paying a stranger's "agent" before seeing the car is a well-known online car-sale scam. Only D meets every requirement. The base model recommended B; the fine-tuned model flagged B as a scam and recommended D.

Appendix: submitted files and how to reproduce

Everything is in the `car_advisor/` folder submitted on Canvas, except that a smaller version of the Canvas zip leaves out the trained adapter (it then contains `ADAPTER_NOTE.txt`). The same files, always including the adapter, are in the GitHub repository, where the adapter is stored as two zip parts that `setup.sh` rebuilds:
https://github.com/rajdeeepm/ai-car-advisor
The `README.md` has full instructions, and the file names below link to the repository.

File	Contents
submission/study_dataset.csv	**Study dataset, 200 trials**: buyer profile and listings A–D (input); model choice, red flags, reason, trade-offs and raw output (model output); per-listing issues, qualifying options, acceptable and best answers, correctness and error category (evaluation); base-model columns for comparison
submission/participant_trials.csv	Participant view for Assignment 3 (no answer key)
data/train.jsonl, data/val.jsonl	Training (1,620) and validation (180) data
data/study.jsonl, data/study_v1.jsonl	Final study set and the iteration-1 version
gen_dataset.py, natural_text.py, make_study_v2.py	Dataset generation
prompts.py, train_lora.py, infer.py, demo.py, run_all.sh	Model training and inference
evaluate.py, make_report_assets.py, export_csv.py	Metrics, error analysis, report tables and figures
results/ft_error_root_causes.csv	Hand-labeled root cause of each fine-tuned error
preds/, results/	Saved raw model outputs and scores for both iterations
model/lora_adapter/, model/log_history.json, setup.sh	Trained LoRA adapter (split zip in the repository; `setup.sh` rebuilds it) and training log
report/report.tex	Source of this report

Reproducing the reported numbers (no GPU, seconds): These commands rebuild every table, figure and number in this report from the saved model outputs:

```
python evaluate.py --preds preds/base_study.jsonl preds/ft_study.jsonl
python make_report_assets.py
```

Reproducing everything: The first command regenerates all data byte-for-byte from the seeds; the second retrains the adapter and regenerates all predictions on one GPU (about 40 minutes):

```
python gen_dataset.py --out data && cp data/study.jsonl data/study_v1.jsonl && python make_study_v2.py
MODEL=<path to Qwen3-4B-Instruct-2507> bash run_all.sh
```

References

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[4] Z. Buçinca, M. B. Malaya, K. Z. Gajos. *To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making.* CSCW, 2021.

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